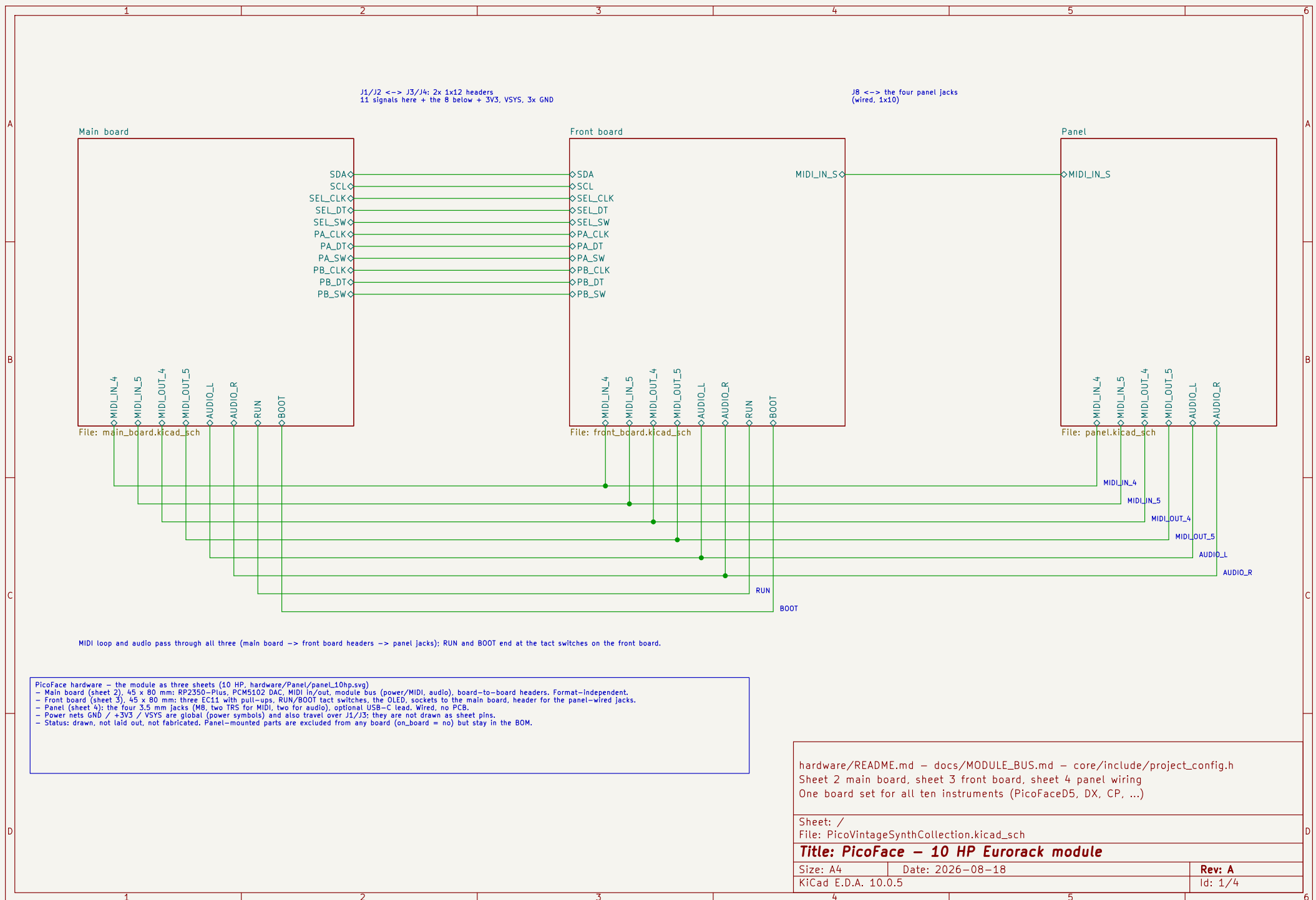
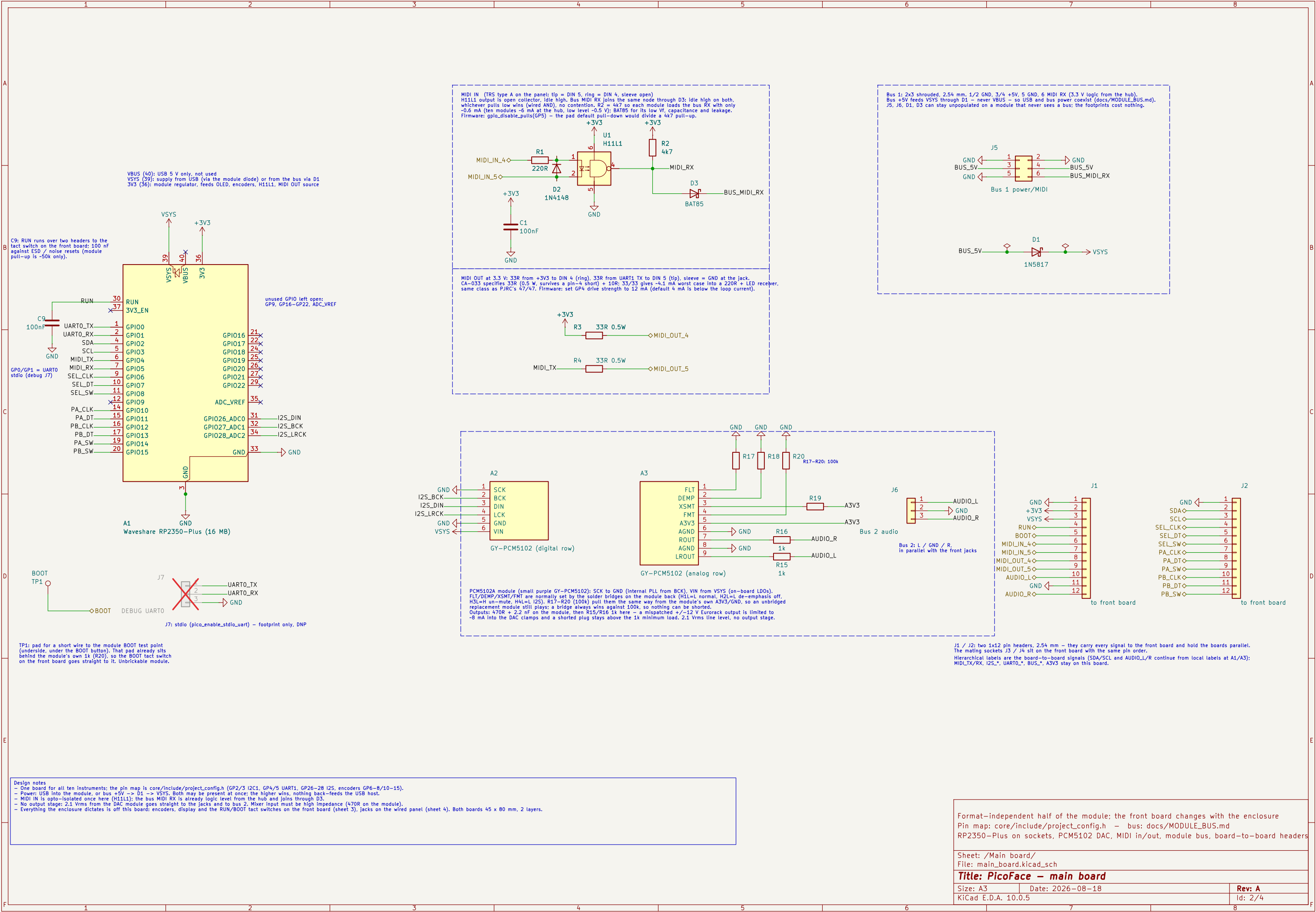
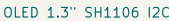








EC11: A/B quadrature to PIO (CLK/DT), switch to a GPIO. 10k pull-ups to +3V3 (the RP2350 internal ~50k are enabled too); 10 nF debounce footprints on A/B – buy, leave unpopulated (the PIO decoder does not need them).



OLED: I2C1 at 1 MHz, u8g2 SH1106 driver, address 0x3C.
Modules carry their own SDA/SCL pull-ups (~4k7).
Pin order GND-VCC-SCL-SDA is the common one – check the module, some vendors swap the first two.
Panel window is cut to the pixels (active area starts 7.35 mm below the module top edge).



J8: 1x10 header, wired to the panel jacks (sheet 4, M8 panel jacks, solder lugs). MIDI IN sleeve has no DC path to ground (CA-033); C8 (DNP) is the optional 100 nF RF ground the spec allows. Two jacks are TRS (MIDI), two are TS or TRS with the ring open (audio).



SW4 RUN / SW5 BOOT: 6x6 mm tact switches, 4-pin (SW_PUSH_6mm), each to GND.
The board sits 6.5 mm behind a 1.6 mm panel, so choose an actuator height of about 9.5 mm or more (H9.5 / H13 types) to reach through the 4 mm panel holes.
RUN: module pull-up + C9 on the main board; BOOT: module pad via TP1 on the main board.

A different enclosure means a new front board; the main board does not change
Board sits 6.5 mm behind the panel (EC11 body height) – hardware/README.md
10 HP: three EC11, RUN/BOOT tact switches, 1.3" SH1106 OLED on standoffs, header for the panel-w

Sheet: /Front board/
File: front_board.kicad_sch

Title: PicoFace – front board

Size: A3	Date: 2026-08-18
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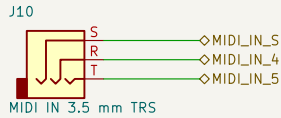
KiCad E.D.A. 10.0.5

[illegible]

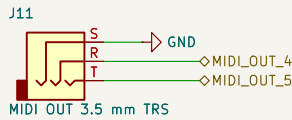
Rev: A

Id: 3/4

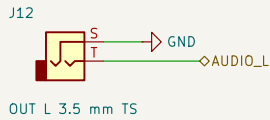
F



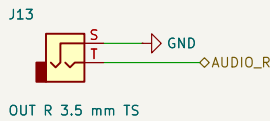
MIDI IN: tip -> J8.1, ring -> J8.2, sleeve -> J8.3
(open: CB on the front board is the optional RF cap)



MIDI OUT: tip -> J8.4, ring -> J8.5, sleeve -> J8.6 (GND)



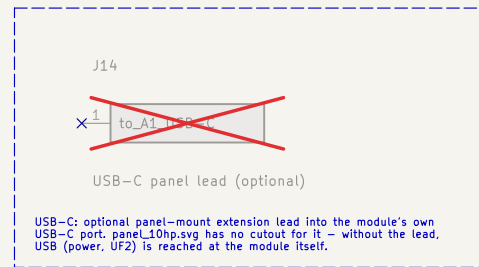
OUT L: tip -> J8.7, sleeve -> J8.8 (GND)



OUT R: tip -> J8.9, sleeve -> J8.10 (GND)

For TRS type B controllers swap tip and ring at the Jack -
It is a wire, not a board change.

Jacks: 3.5 mm, M8 thread, 10 mm dia, nut, solder lugs - stereo for MIDI (3 poles),
mono or stereo for audio (ring left open). A metal bushing ties the sleeve to a metal panel:
fine for OUT/L/R (GND), for MIDI IN use an insulating washer or accept it (the opto still isolates).



Panel - 10 HP, 50.8 x 128.5 mm, 3U (hardware/Panel/panel_10hp.svg)
- The four jacks mount to the 1.6 mm panel (M8 nut) and wire to J8 on the front board. Sleeves of MIDI OUT and both
audio jacks are GND (J8.6/8/10); the MIDI IN sleeve stays open (CB = optional 100 nF).
- Jack row at panel Y 109.6, X 8.15 / 19.65 / 31.15 / 42.65 (MIDI IN, MIDI OUT, L, R) - below the 45 x 80 front board.
- RUN and BOOT are 6x6 tact switches on the front board now (sheet 3), reached through 4 mm holes in the panel.

Drill list: hardware/README.md
TRS MIDI type A: tip = DIN 5, ring = DIN 4, sleeve = DIN 2 (shield)
Panel jacks wired to J8 on the front board; 10 HP panel: hardware/Panel/panel_10hp.svg

Sheet: /Panel/
File: panel.kicad_sch

Title: PicoFace - panel (wired parts)

Size: A4
KiCad E.D.A. 10.0.5

Date: 2026-08-18

Rev: A
Id: 4/4